EXP 13: Case Study On Low-Code Limitations

AIM

To study the limitations of low-code development platforms using Zoho Creator and analyze their impact on application development, customization, performance, integration, and scalability.

PROBLEM STATEMENT

A financial institution wants to develop a Loan Application Management System using Zoho Creator. The application should manage applicant details, loan requests, document verification, credit checks, approval processing, notifications, and reports.

Zoho Creator enables rapid application development through visual tools, predefined components, workflows, and Deluge scripting. However, as business requirements become more complex, certain limitations of low-code development may affect flexibility and scalability.

SYSTEM REQUIREMENTS

- Applicant registration and loan application submission.
- Loan type and requested amount management.
- Document verification and credit checks.
- Loan approval and rejection processing.
- Email and SMS notifications.
- Reports and dashboards.
- Integration with external banking and credit verification systems.

PROCEDURE

1. Log in to Zoho Creator and open the Loan Application Management System.

2. Create or review forms for applicant details, loan information, document verification, and approval status using the drag-and-drop interface.

3. Configure or examine workflows and business rules using Deluge scripting.

4. Identify limitations in advanced customization, complex logic, third-party integrations, and UI design.

5. Analyze application performance requirements as the number of loan records and users increases.

6. Examine integration requirements and platform-specific dependencies.

7. Analyze the limitations and identify suitable workarounds or alternative technologies.

8. Document the observations and conclude the suitability of low-code development for the Loan Application Management System.

LIMITATIONS IDENTIFIED

1. Limited Customization

Low-code platforms provide predefined UI components and templates. Creating a completely unique loan application interface or implementing highly customized user interactions may require workarounds or external development.

2. Complex Business Logic

Simple conditions and workflows are easy to implement. However, complex loan eligibility calculations, multiple approval dependencies, credit risk evaluation, and advanced financial algorithms may require extensive Deluge scripting or external services.

3. Platform Dependency (Vendor Lock-in)

Applications developed using Zoho Creator depend on its platform-specific tools, data models, and Deluge language. Migrating the Loan Application Management System to another platform may require rewriting scripts and redesigning workflows.

4. Performance and Scalability

As the number of loan applications, applicants, and reports increases, inefficient queries, excessive workflow execution, or platform resource limits may affect performance. Large-scale financial applications require careful optimization and testing.

5. Integration Limitations

Integrating external banking systems, credit bureaus, payment gateways, or third-party APIs may require API configuration, authentication, custom functions, and additional development.

6. Limited Control Over Infrastructure

Zoho Creator manages much of the underlying infrastructure. Developers have less direct control over server configuration, database optimization, deployment architecture, and low-level system settings than with a fully custom application.

7. Security and Compliance Considerations

Although Zoho Creator provides security features, developers must correctly configure permissions, record access, authentication, and data sharing.

Sensitive applicant information, such as PAN numbers, residential addresses, income details, and credit scores, requires appropriate protection. Financial-sector compliance requirements may need additional controls and verification.

8. Licensing and Cost

Advanced features, higher usage, additional users, or greater resource requirements may affect subscription costs. Organizations should evaluate pricing and plan limits before deploying large-scale loan management applications.

For example, advanced deployment environment features may require a paid Zoho Creator subscription.

PROPOSED SOLUTION

Limitation	Proposed Solution
Limited customization	Use custom pages, supported HTML/CSS, integrations where appropriate
Complex business logic	Use optimized Deluge scripts or external services
Vendor lock-in	Document workflows and maintain exportable data backups
Performance issues	Optimize loan reports, queries, and workflows
Integration challenges	Use supported APIs, webhooks, and connectors
Infrastructure restriction	Use external services when greater control is required
Security concerns	Apply least-privilege permissions and regularly audit access
Cost concerns	Review usage, licensing, and resource requirements

OBSERVATION

S. No.	Observation	Result
1.	Forms created using drag-and-drop tools	Faster development.
2	Basic workflows configured	Less manual coding.
3	Complex logic examined	Additional scripting may be required.
4	External integrations examined	API configuration may be necessary.
5	Performance requirements analyzed	Optimization becomes important as usage grows.
6	Security permissions configured	Access can be controlled through appropriate settings.
7	Platform dependency evaluated	Migration may require redevelopment.
8	Licensing requirements reviewed	Cost planning is necessary.

RESULT

The limitations of low-code development in Zoho Creator were studied through the Loan Application Management System case study. The analysis identified challenges related to customization, complex business logic, integration, performance, scalability, security, cost, and platform dependency. Suitable solutions were proposed to reduce their impact.

CONCLUSION

Thus, the case study on low-code limitations in Zoho Creator was completed successfully. Low-code platforms enable rapid application development, reduce manual coding, and simplify workflow automation.

However, they may impose restrictions on customization, infrastructure control, complex logic, and portability. Therefore, developers should evaluate application requirements, expected workload, integration needs, security, and long-term costs before choosing a low-code platform.

For complex loan processing requirements, combining Zoho Creator with custom code and external services can provide a practical solution.